Linear Equations & Systems

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

If $2(x-6)=3x-5$, what is the value of x?

2 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

$\{x+6y=-43;\ -x-y=13$ What is the value of y?

3 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

$\{6x-3y=-6;\ -6x+5y=-2$ What is the value of y?

4 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

If $3(x-2)=-3x-12$, what is the value of x?

5 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

If $2x+5=-5x+47$, what is the value of x?

6 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

If $6(x+1)=-x+20$, what is the value of x?

7 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

What is the sum of all solutions to $|3x-9|=15$?

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

If $2(x+7)=3x+19$, what is the value of x?

9 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

If $2(x-5)=4x-12$, what is the value of x?

10 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

$\{2x-2y=8;\ -2x+3y=-7$ What is the value of y?

11 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

If $2(x+1)=3x-1$, what is the value of x?

© ClipSAT — clipsat.org | All Rights Reserved

12 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

If $8x - 7 = -2x + 73$, what is the value of x?

13 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

$\{2x + 6y = -48;\ -2x + 3y = -33$ What is the value of x?

14 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

$\{5x - 5y = -10;\ -5x - 4y = 64$ What is the value of $x + y$?

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

$\{2x - 5y = -9;\ -2x - 2y = 16$ What is the value of y?

16 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

If $5x - 10 = 4x - 6$, what is the value of x?

17 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

$\{x + 5y = -27;\ -x - 2y = 9$ What is the value of $x + y$?

18 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

$\{x - y = -1;\ -x + 3y = -1$ What is the value of x?

19 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

$\{x - 4y = -34;\ -x + 6y = 50$ What is the value of $x + y$?

20 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

Consider the system: $\{3x + 2y = 19; 2x - y = 8$ What is the value of $x + y$?

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

21 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $3x + 14 = 32$, what is the value of x?

a. 15.333333333333334

b. 18

c. 6

d. 7

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $2(x - 8) = 8$, what is the value of x?

a. 0

b. 12

c. 4

d. -4

23 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

A moving company charges a one-time fee of USD 30 plus USD 55 per mile. If the total charge was USD 305, how many miles was the move?

a. 5

b. 6

c. 4

d. 7

24 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $\frac{x}{3} + 5 = 13$, what is the value of x?

a. 27

b. 24

c. 8

d. 39

25 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $8x + 12 = 4x + 36$, what is the value of x?

a. 6

b. 2

c. 24

d. 8

26 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $5x = 15$, what is the value of $10x$?

a. 30

b. 3

c. 15

d. 6

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

A taxi company charges a one-time fee of USD 25 plus USD 30 per mile. If the total charge was USD 175, how many miles was the trip?

a. 6

b. 4

c. 7

d. 5

28 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $\frac{x}{2} + 3 = 9$, what is the value of x?

a. 12

b. 6

c. 18

d. 14

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $7x - 8 = 8x - 15$, what is the value of x?

a. 11

b. 3

c. 7

d. 13

30 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

$\{5x + y = -49;\ -5x - 4y = 76$ What is the value of $x + y$?

a. -18

b. -17

c. -10

d. -9

31 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $8x + 1 = -x - 26$, what is the value of x?

a. 4

b. 3

c. -3

d. -10

© ClipSAT — clipsat.org | All Rights Reserved

32 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $8x + 2 = -4x - 58$, what is the value of x?

a. -5

b. -1

c. 4

d. -10

33 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $7x - 2 = -8x + 28$, what is the value of x?

a. 3

b. 1

c. -3

d. 2

34 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

$\{x + 5y = 24;\ -x + 2y = 18$ What is the value of $x + y$?

a. 7

b. 3

c. 0

d. -8

35 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

$\{6x - 6y = 12;\ -6x - 2y = 12$ What is the value of x?

a. -1

b. -3

c. -5

d. -6

36 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $\frac{2}{x} + \frac{3}{2x} = \frac{7}{12}$, what is the value of x?

a. 4

b. 5

c. 6

d. 7

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

A coffee shop sells small coffees for USD 2.50 and large coffees for USD 4.00. On Monday, the shop sold 80 total coffees for USD 260. How many large coffees were sold?

a. 35

b. 40

c. 45

d. 50

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

$\{4x + 5y = 13;\ -x + 6y = 4$ What is the value of x?

a. 9

b. 2

c. 0

d. 11

39 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

$\{4x + 6y = 4; 12x + ky = -15$ For what value of k does this system of equations have no solution?

a. 20

b. 21

c. 18

d. 13

40 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

$\{4x + 2y = 2; 8x + ky = -10$ For what value of k does this system of equations have no solution?

a. 9

b. 4

c. 10

d. 5

41 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

$\{6x + 5y = 4;\ -2x - y = -4$ What is the value of x?

a. 4

b. 2

c. 9

d. 10

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

What is the sum of all solutions to the equation $|2x + 2| = 10$?

a. -5

b. -10

c. -2

d. -3

43 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $2x + 3y = 14$ and $5x + 4y = 28$, what is the value of $7x + 7y$?

a. 38

b. 42

c. 46

d. 52

44 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

For what value of p does the equation $4(2x - 3) + p = 8x - 5$ have infinitely many solutions?

a. -7

b. -5

c. 5

d. 7

45 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

A chemist needs to mix a 20% acid solution with a 50% acid solution to produce 60 liters of a 35% acid solution. How many liters of the 50% solution are required?

a. 25

b. 30

c. 35

d. 40

© ClipSAT — clipsat.org | All Rights Reserved